Summary in Graph

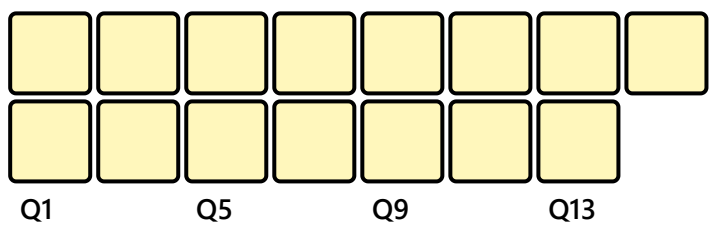
# Exam Summary (GO Classes CS Test Series 2026 | Theory of Computation | Topic Wise Test 3)

| | | | |
|---|---|---|---|
| Qs. Attempted: | 0<br>0 + 0 | Correct Marks: | 0<br>0 + 0 |
| Correct Attempts: | 0<br>0 + 0 | Penalty Marks: | 0<br>0 + 0 |
| Incorrect Attempts: | 0<br>0 + 0 | Resultant Marks: | 0<br>0 + 0 |

| | |
|---|---|
| Total Questions: | 15<br>5 + 10 |
| Total Marks: | 25<br>5 + 20 |
| Exam Duration: | 45 Minutes |
| Time Taken: | 0 Minutes |

## Individual Question Statistics

Select Metric: Marks Obtained




* Average Stats based on first-attempt stats of Top 10% (115) out of 1152 users

EXAM RESPONSE EXAM STATS FEEDBACK

# Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Theory of Computation

Consider the following language $\mathrm{L}$ with the alphabet $\{0, 1\}$. The empty string is not in $\mathrm{L}$. If the string is nonempty and has even length, then it is in $\mathrm{L}$. If the string has odd length, then it is in $\mathrm{L}$ if and only if it is a palindrome.

Which of the following best describes $\mathrm{L}$?

A. Regular

B. DCFL but not regular
C. CFL but not DCFL
D. Not CFL

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Consider the following languages :

i. $\mathrm{L}_1 = \{ww \mid w \in \{a\}^*\}$
ii. $\mathrm{L}_2 = \{ww \mid w \in \{a, b\}^*\}$

Which of the above languages is Non-regular But CFL?

A. Only $\mathrm{L}_1$
B. Only $\mathrm{L}_2$
C. Both
D. None

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

Consider the following languages :

1. The language of the $\mathrm{CFG}$ with productions $\mathrm{S} \rightarrow \mathrm{AS} \mid \mathrm{SB} \mid \epsilon$, $\mathrm{A} \rightarrow 0$, and $\mathrm{B} \rightarrow 1$
2. The language of regular expression $0^*1^*$

Which of the following is true ?

A. $1$ is proper subset of $2$
B. $2$ is proper subset of $1$
C. $1 = 2$
D. Neither $1$ is subset of $2$, nor $2$ is subset of $1$.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Theory of Computation

Which of the following languages are $\mathrm{CFL}$ But Not Regular?

A. $\mathrm{L} = \left\{a^{2n}b^{3n} \mid \mathrm{n} > 0\right\}$
B. $\mathrm{L} = \left\{a^{2n}b^{3m} \mid \mathrm{m}, \mathrm{n} > 0\right\}$
C. $\mathrm{L} = \left\{a^{2n}b^{3} \mid \mathrm{n} > 0\right\}$
D. $\mathrm{L} = \left\{a^{2n}b^{n+m} \mid \mathrm{n}, \mathrm{m} > 0\right\}$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Compiler Design

A grammar is said to be "useless" if and only if it produces no terminal strings. If $\mathrm{S}$ is the start symbol, which of the following grammars is (are) "useless"?

I. $\mathrm{S} \rightarrow \mathrm{AB}|\mathrm{AS}$
$\mathrm{A} \rightarrow \mathrm{B} \mid a$
$\mathrm{B} \rightarrow \mathrm{A} \mid b$

II. $\mathrm{S} \rightarrow \mathrm{SA} \mid \mathrm{AS}|\mathrm{SB}$
$\mathrm{A} \rightarrow a$
$\mathrm{B} \rightarrow a \mid b$

III. $\mathrm{S} \rightarrow \epsilon|\mathrm{A}$
$\mathrm{A}\rightarrow \mathrm{B}$
$\mathrm{B}\rightarrow \mathrm{C}$
$\mathrm{C} \rightarrow a$

A. None
B. Only II
C. Only III
D. Only II, III

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Consider the following Grammar:

$$\mathrm{S} \rightarrow rl \mid r\mathrm{S}l \mid \mathrm{SS}$$

Which of the following strings can be generated by this grammar?

A. $rrllrl$
B. $rllrrrll$
C. $rrlrl\ rllrl\ rrl$
D. $rrrll\ lrlr\ rll$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 05sec | Discuss

Q #7 | Multiple Select Type | Award: 2 | Penalty: 0 | Compiler Design

Which of the following statements are true?

A. The grammar, $\mathrm{G} : \mathrm{E} \rightarrow \mathrm{E}+\mathrm{E} \mid \mathrm{E}*\mathrm{E} \mid x$ is ambiguous.
B. A regular grammar can never be ambiguous.
C. A grammar is ambiguous if there is some string $w$ such that $w's$ right-most derivation differs from its left-most derivation.
D. A grammar is ambiguous if there is some string w such that w has two different parse trees.

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Let $\mathrm{L} = \{a^p \mid p \text{ is prime}\}$ be a language. Which of the following are true?

A. $\mathrm{L}$ is Not context free language.
B. $\mathrm{L}^*$ is Not Context free language.
C. Complement of $\mathrm{L}^*$ is regular.
D. Complement of $\mathrm{L}$ is $\mathrm{CFL}$.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

A context-free grammar has a set of terminals $\{0, 1, -\}$, a set of non-terminals $\{\mathrm{N}, \mathrm{P}\}$, where $\mathrm{N}$ is the start symbol, and productions given by the following .

$$\mathrm{N} \rightarrow 0 \mid \mathrm{P} \mid - \mathrm{P}$$
$$\mathrm{P} \rightarrow 1 \mid \mathrm{P}\ 0 \mid \mathrm{P}\ 1$$

What can be said about the language generated by this grammar?

A. It is regular.
B. It is Non-regular But DCFL.
C. It is non-DCFL but CFL.
D. It is Non-CFL.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following languages are not context free?

A. $\mathrm{L}_1 = \{0^n 1^{2n} \mid n \geq 0\}$
B. $\mathrm{L}_2 = \{w \in \{0,1\}^* \mid w \neq w^{\mathrm{R}}\}$.
C. $\mathrm{L}_3 = \{www \mid w \in \{0,1\}^*\}$.
D. $\mathrm{L}_4 = \{x/y/z \mid x, y, z \in \{0,1\}^*$ are binary numbers such that $x + y = z\}$. The alphabet for this language is $\{0, 1, /\}$. For example, $10/10/100 \in \mathrm{L}_3$ and $11/1/001 \notin \mathrm{L}_3$.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Every day, a weather station records whether the day was sunny $(\mathrm{S})$, cloudy $(\mathrm{C})$ or rainy $(\mathrm{R})$. A sequence of records over several days is a string in $\{\mathrm{S}, \mathrm{C}, \mathrm{R}\}^*$. We call a sequence of records $\mathrm{GO}$-like if:

- There are never more than three consecutive rainy days, and
- There are never more than five consecutive non-sunny days.

The language of $\mathrm{GO}$-like sequences is

A. Regular
B. DCFL but Not regular
C. CFL but not DCFL

D. Not CFL

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

Consider the following languages:

1. The language (accepted by final state) of $\mathrm{PDA}(\{p,q\},\{0,1\},\{\mathrm{X},\mathrm{Z}\},\delta,q,\mathrm{Z},\{p\})$ with $\delta$ defined by the rules $\delta(q,0,\mathrm{Z})=\{(q,\mathrm{X}\ \mathrm{Z})\},\delta(q,0,X)=\{(q,\mathrm{X}\ \mathrm{X})\},\delta(q,1,\mathrm{X})=\{(p,\epsilon)\}$, and $\delta(p,1,\mathrm{X})=\{(p,\epsilon)\}$.
2. The language of the CFG with productions $\mathrm{S}\rightarrow 0\ \mathrm{S}\ 1 \mid 0\ \mathrm{S} \mid \epsilon$.

Which of the following is true ?

A. $1$ is proper subset of $2$
B. $2$ is proper subset of $1$
C. $1=2$
D. Neither $1$ is subset of $2$, nor $2$ is subset of $1$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following languages is/are Context Free(CFL)?

A. $\{w\#w' \mid w,w' \in \{a,b\}^* \text{ and } w \neq w'\}$.
B. $\{0^n1^m \mid n \leq m \leq 2n\}$.
C. $a^*b^*c^* - \{a^nb^nc^n \mid n \geq 0\}$.
D. $\{w \in \{0,1,2\}^* \mid \#_0(w)=\#_1(w)=\#_2(w)\}$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

An infinite language is $\mathrm{REG\text{-}immune}$ if it contains no infinite subset that is a regular language.

Which of the following are $\mathrm{REG\text{-}immune}$ languages?

A. $\{a^nb^n \mid n \geq 0\}$
B. $\{ww \mid w \in (a+b)^*\}$
C. $\{w\#w \mid w \in (a+b)^*\}$
D. $\{a^p \mid p \text{ is prime}\}$

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | Compiler Design

$$\begin{aligned}\mathrm{S} &\rightarrow \mathrm{A0B}\\ \mathrm{A} &\rightarrow \mathrm{BB} \mid 0\\ \mathrm{B} &\rightarrow \mathrm{AA} \mid 1\end{aligned}$$

What is the number of terminal strings of length $5$ generated by the context-free grammar shown above?

**Your Answer:** | **Correct Answer: 5** | Not Attempted | Time taken: 00min 00sec | Discuss

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